

Improving the modulation bandwidth of LED by CdSe/ZnS quantum dots for visible light communication

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Abstract: Visible light communication (VLC) is an advanced and high-efficiency wireless communication technology. As one of the most important light sources in VLC, conventional white light emitting diode (WLED) based on $\text{Y}_3\text{Al}_5\text{O}_{12}:\text{Ce}^{3+}$ (YAG:Ce) phosphor limits the system transmitting rate severely due to its narrow modulation bandwidth. Considering the short fluorescent lifetime of quantum dots (QDs), QD-LEDs with wide modulation bandwidths were designed here to improve the transmitting rate of VLC. CdSe/ZnS core/shell QDs and related luminescent microspheres (LMS) were implemented as light conversion materials for the QD-LEDs. Compared with conventional phosphor WLED, the proposed QD-LED and QD-WLED reached maximum improvement on modulation bandwidth at 74.19% and 67.75% respectively. Furthermore, mathematical modeling of smearing was analyzed to establish the relationship between fluorescent lifetime and modulation bandwidth. Our findings will provide an effective solution of white LEDs for high speed VLC.

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1. Introduction

Visible light communication (VLC) is an advanced wireless communication technology that can realize LED lighting and optical communications together by modulating LED light source [1]. Considering its advantages including high transmitting rate, low energy consumption and enhanced security, this technology has garnered a lot of attention these years in data communications [2,3]. However, as one of the most important light sources in VLC, conventional white LED (WLED) based on $\text{Y}_3\text{Al}_5\text{O}_{12}:\text{Ce}^{3+}$ (YAG:Ce) phosphor has long fluorescent lifetime, which results in a narrow modulation bandwidth and, as a consequence, the system transmitting rate has been severely limited [4]. Therefore, how to increase the bandwidth of the white light source has become a key issue.

In recent years, many studies have been performed on this problem to make further improvement of the transmitting rate for VLC system. Normally, there are mainly three methods reported in previous literatures, which could be summarized as following: Optimizing the structure of LED chips, using blue-filtering and utilizing equalization circuit as well. Optimizations on the LED chip structures aim at reducing the Resistance-Capacitance (RC) time constant and recombination time of carriers, and consequently, improving the bandwidth of blue LED chips [5–7]. Blue-filtering is always applied to indoor VLC with WLED, which could efficiently remove the undesired signals produced by yellow phosphor [8,9]. In addition, equalization circuit could reduce the signal distortions by resistor, inductance and resonant capacitor [10–12]. According to these researches, optimizations on structure of LED and external accessories are mainly considered. On the other hand, as one of the most important parts in WLED, traditional light conversion material (e.g. commercial YAG:Ce phosphor) narrows down the bandwidth Prominently and this problem need to be solved urgently.

Nowadays, quantum dots (QDs) have emerged as promising materials in lighting as well as display due to their tunable emission wavelength, narrow full width at half maximum (FWHM), nearly perfect photoluminescence (PL) and high quantum yield [13–16]. In particular, as nanocrystals, QDs could well confine the excitons in narrow distributed recombination channels, as a result of that, QDs have short fluorescent lifetime, which is far less than that of commercial YAG:Ce phosphor [17]. For light source, therefore, QD-LED could improve not only the color quality but also the bandwidth of WLED to obtain a better performance in VLC. On this issue, reference [17] reported a QDs/polyimide nanocomposite

system to improve the modulation bandwidth of white LED based on micro-LED chip. This work created a possible path to integrated hybrid visible light sources for communications applications. Nevertheless, for QDs solution, interferences, such as defects, scattering caused by the polymer protection layers, etc., are less than those in QDs composite, so that disturbances on modulation characteristics in QDs solution will be weaker than that in QD composites [15]. Therefore, it is necessary to measure modulation bandwidth of QDs solution to seek out intrinsic modulation properties of QDs. However, the effects of QDs solution on, and the difference between QDs solution and QDs composites in, improving the modulation bandwidth of LED have not been mentioned in reference [17]. Moreover, types of LED chips (e.g. micro-LED, commercial LED with large size, etc.) also have great effects on the modulation bandwidth of WLED. To minimize instrumental error effectively, one same LED chip should be adopted in different QD-LEDs, QD-WLEDs and phosphor WLEDs.

In this work, we obtained CdSe/ZnS core/shell QDs solution and composites with different emission peak including 525nm as well as 625nm to fabricate green QD-LED, red QD-LED and more importantly, QD-WLED, which can be greatly practical in VLC. According to experimental results, QD-LED presents wider modulation bandwidth than conventional WLED with YAG:Ce phosphor. Furthermore, mathematical modeling of the smearing has been analyzed. Combining analysis results with fluorescent lifetime, it can be found that the red QDs, which have a shorter lifetime than green QDs, leads to a higher bandwidth improvement. To the best of our knowledge, this is the first time the relationship between QDs' fluorescent lifetime and bandwidth of LED is explicitly analyzed.

2. Experiment

2.1 Materials preparation

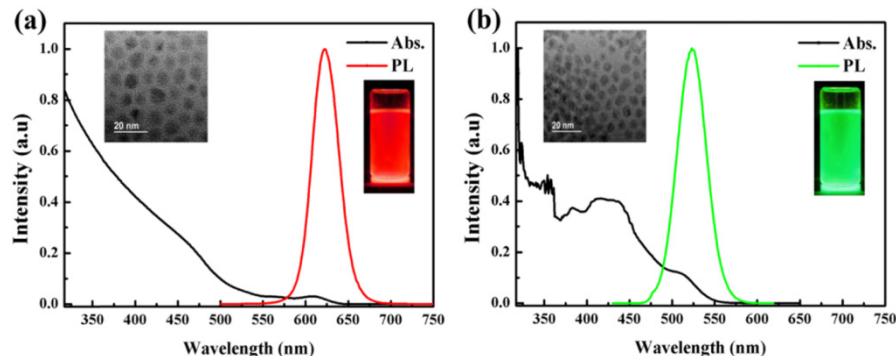


Fig. 1. (a) Spectra of red QDs. (Insert: Red QDs solution sample and high-resolution transmission electron microscopy (HRTEM) of red QDs). (b) Spectra of green QDs. (Insert: Green QDs solution sample and HRTEM of green QDs).

In this work, we adopted red and green CdSe/ZnS core/shell QDs as light conversion materials. High quality red CdSe/ZnS core-shell QDs were prepared by the modified tri-n-octylphosphine-assisted successive ionic layer adsorption and reaction (TOP-SILAR) method described in our previous research [15, 16]. Figure 1(a) and 1(b) show the absorption spectra and the PL emission spectra of red and green QDs respectively. The emission peak is 623 nm for red QDs and 523 nm for green QDs. Meanwhile, the FWHM is 36 nm and 40 nm for red and green QDs respectively. In addition, as shown in the high-resolution transmission electron microscopy (HRTEM) images in the inserted picture of Fig. 1, red and green QDs could be observed as small spheres with 5 nm and 4 nm average diameters separately. Moreover, all the measured nanoparticles demonstrate uniform size distribution.

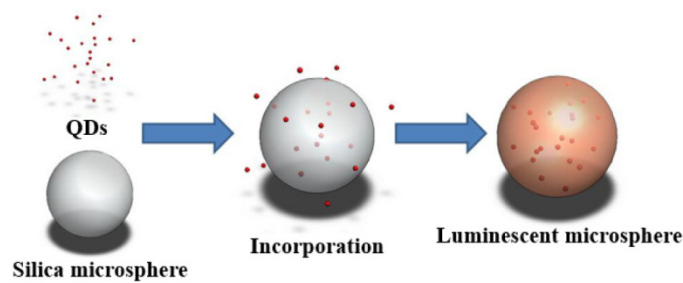


Fig. 2. Schematic of obtaining QDs luminescent microspheres.

To improve the operational performance of QDs and maintain large extent luminescent properties of QDs in WLED, luminescent microspheres (LMS) have been fabricated. To fabricate LMS, 100mg of silica powder with pore size of about 7 nm and diameter in 30 to 60 μm were dispersed in 20 mL n-hexane. The compound solvent was heated and maintained at 60°C with rapid stirring in an open single neck flask for about 2 hrs. As shown in Fig. 2, during the heated process, solvent of n-hexane was intermittently injected to the flask in case of desertification before CdSe/ZnS QDs incorporating into the silica lattice as many as possible. After swelling process, the solvent was evaporated completely to obtain LMS powders [15, 16]. Owing to the lattice structure of LMS, QDs were effectively protected from intrinsic aggregation in matrix and surface deterioration by encapsulant, oxygen and moisture. The LMS composite has therefore maintained large extent luminescent properties of QDs, especially for the high quantum efficiency.

2.2 LED modules fabrication

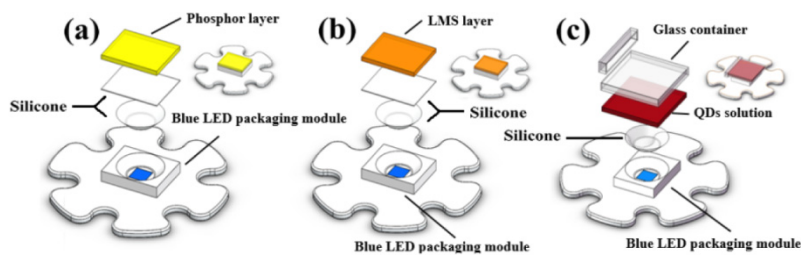


Fig. 3. The schematic of LED modules based on (a) YAG:Ce phosphor; (b) LMS; and (c) QDs solution.

LED modules based on YAG:Ce phosphor, LMS and red or green QDs solution have been designed in our work. Firstly, a blue LED packaging module, which was consist of leadframe, LED chip and related components, was mounted on the substrate. Then, silicone layers mixed with LMS or YAG:Ce phosphor were fabricated. In addition, a thin-walled glass container with QDs solution in it was used to achieve QDs layers. As is shown in Fig. 3(a), (b) and (c), to fabricate LEDs, layers based on phosphor, LMS and QDs solution could be mounted on the same blue LED packaging module by silicone to minimize instrumental error effectively.

2.3 Measurement

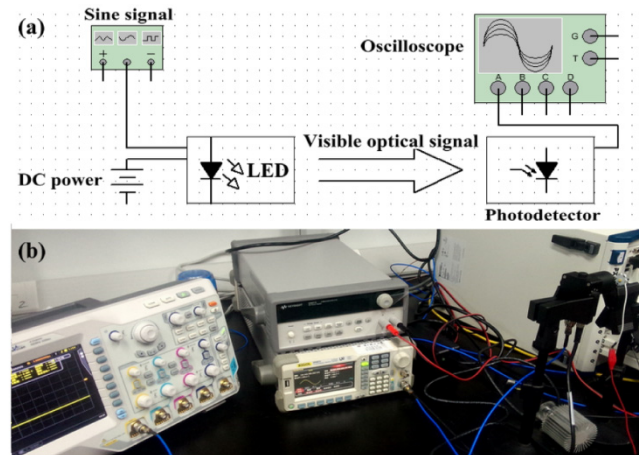


Fig. 4. (a) Schematic of bandwidth measuring. (b) Bandwidth measuring system.

A bandwidth measuring system has been set up as shown in Fig. 4(a) and (b), which includes a direct current (DC) stabilized power supply, a signal generator (Type: RIGOL DG5072) to achieve bias voltage including DC voltage shift and sine signal, a DC-coupled Si transimpedance amplified photodetector (THORLABS PDA10A) with bandwidth of 150 MHz and an oscilloscope. To reduce the instrumental error, firstly DC power was applied on each LED so optical signals sent by these LEDs would be received by photodetector. Adjusting the working voltage for every LED until photodetector generated the same current. Then, for bandwidth measuring, signal generator produced a sine wave output with those above-mentioned voltages as DC bias. While signal generator was producing signals and LED was sending them, photoelectric detector could receive and show them in oscilloscope. Initially, the frequency of sine wave was set at 100 KHz and the peak-to-peak voltage was set at 500mV. After that, the peak-to-peak value of the sine wave was decreased as we increased the frequency of the sine wave. Once the peak-to-peak value dropped to 0.707 times of its initial value, the corresponding frequency was considered as the 3dB bandwidth of LED.

3. Results and discussions

3.1 Analyses of QD-LED modules based on QDs solution

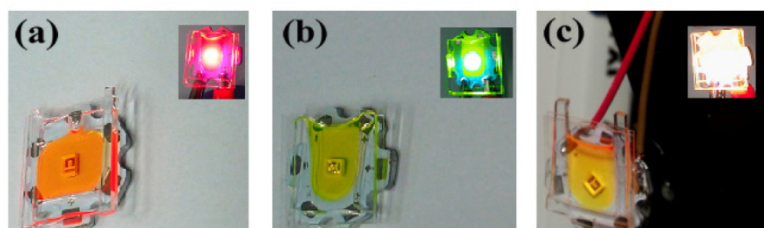


Fig. 5. LED modules with (a) red QDs solution; (b) green QDs solution; and (c) mixture solution of red and green QDs.

Figure 5 shows QD-LED modules based on red QDs solution, green QDs solution and mixture of them. As is shown in Fig. 5(a) and 5(b), red and green light can be observed. For red and green QD-LED, bias voltage and current applied on them were 3.02V and 96mA, and 3.05 V and 107 mA respectively, and luminous efficacies were 12.28lm/W and 74.13lm/W respectively. Meanwhile, QD-WLED, which is showed in Fig. 5(c), can be obtained by

controlling the mixing ratio of red and green QDs. For QD-WLED, bias voltage and current applied on it were 3.03V and 101mA respectively, and luminous efficacy was 54.48lm/W and its CRI is 61.5. It should be noted that the specific LED packaging module we designed mainly focused on realizing convective benchmarking and helping measuring bandwidth more accurately, and the performance of QD-WLED still has room to improve for general lighting in the further [15, 16].

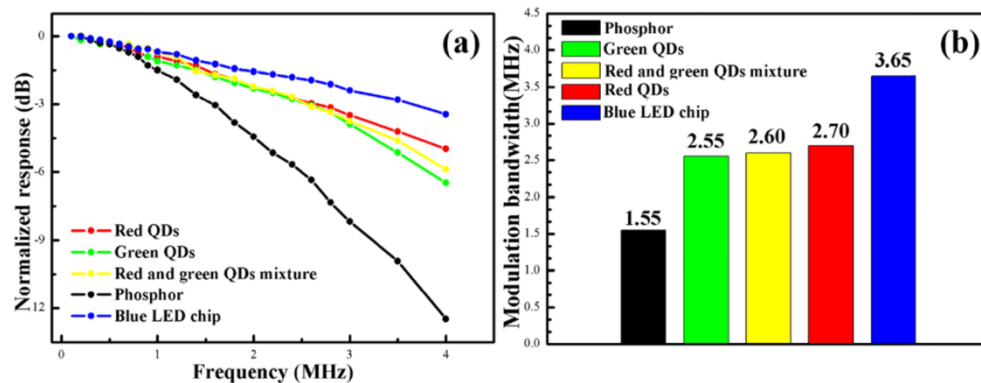


Fig. 6. Comparisons of (a) frequency responses and (b) bandwidths between different LED modules with phosphor, red, green and mixture QDs and without phosphor or QDs.

Figure 6 (a) and (b) show the comparisons of frequency responses and bandwidths between five different LED modules respectively. With the implementation of LEDs based on YAG:Ce phosphor, green QDs, red and green QDs mixture, red QDs and GaN LED chip, the measured 3dB bandwidth results are 1.55 MHz, 2.55MHz, 2.60 MHz, 2.70 MHz and 3.65 MHz respectively. Compared with ordinary YAG:Ce phosphor, the improvement for bandwidth is 74.19% by using red QDs in QD-LED, which is the highest improvement in these samples. Meanwhile, green QD-LED and QD-WLED also have a good performance on bandwidth with improvements of 64.52% and 67.75%, which are far better than phosphor WLED. Furthermore, with the increase of frequency gradually, the normalized responses of QD-LEDs decrease more slowly than that of phosphor WLED. Especially, as is shown in Fig. 6 (a), 4 MHz frequency corresponds to -13dB response for the phosphor WLED while only -6dB response for the QD-WLED. Therefore, compared with ordinary phosphor WLED, the higher the frequency is, the better the modulation characteristic of QD-LED is. Besides, since there is no fluorescent material, GaN LED chip has the best performance on frequency responses and bandwidth reaching 3.65MHz.

3.2 Analyses of LED modules based on phosphor and LMS

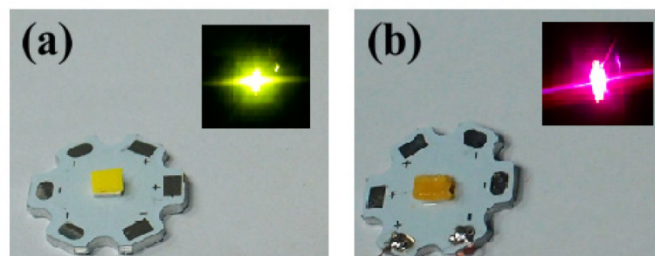


Fig. 7. LED modules with (a) YAG:Ce phosphor layer; and (b) red LMS layer.

Figure 7 (a) and (b) show LED modules based on yellow YAG:Ce phosphor and red LMS layer respectively. In these modules, YAG:Ce phosphor or LMS was mixed with silicone to

fabricate layers. For phosphor based LED module, bias voltage and current applied on it were 2.84V and 54mA respectively, and its luminous efficacy was 119.23lm/W. On the other hand, for the LMS based LED module, bias voltage and current applied on it were 2.98V and 90mA separately, and the luminous efficacy was 23.76lm/W.

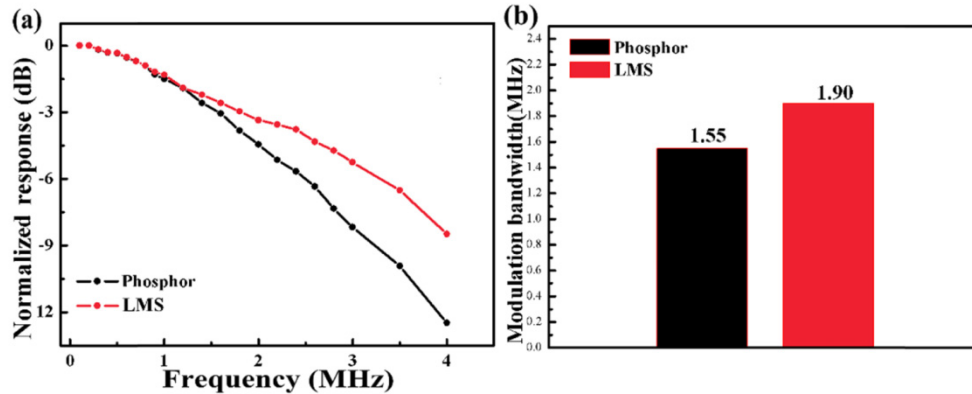


Fig. 8. Comparisons of (a) frequency responses and (b) bandwidths between different LED modules with phosphor and LMS.

Figure 8 (a) and (b) show the comparisons of frequency responses and bandwidths between two different LED modules respectively. The measured 3dB bandwidth results are 1.55 MHz and 1.90MHz for LED modules with YAG:Ce phosphor and LMS respectively. Compared with phosphor, utilizing LMS as light conversion material for LEDs can increase the bandwidth by 22.58%. For LMS, since the defect is formed in the synthetic process, and meanwhile, light scattering is prominent, the light attenuation in LMS will be slower than that in QDs solution. Therefore, the response of LMS is not so sensitive to the change of signals. As a consequence, the 3dB bandwidth of LMS based QD-LED is not as wide as that of QDs solution based. On the other hand, since LMS are solid powders with much better long term operational performance than QDs solution, LMS still have great potential for QD-LED based VLC.

3.3 Analysis on the relation between fluorescent lifetime and bandwidth

Fluorescent lifetime of QDs was measured by FLSP920 Steady/Transient State Fluorescence Spectrometer (Edinburgh Instruments) with the Time Correlated Single Photon Counting (TCSPC) technology. Figure 9 shows that the fluorescent lifetime of red QDs (26.31 ns) is shorter than that of green QDs (29.34 ns). Because of the smearing, light conversion materials limit the bandwidth of LED. On the other hand, fluorescent lifetime of light conversion materials is tightly linked to the generating of smearing. To establish the relation between fluorescent lifetime and bandwidth, mathematical modeling of the smearing has been analyzed here.

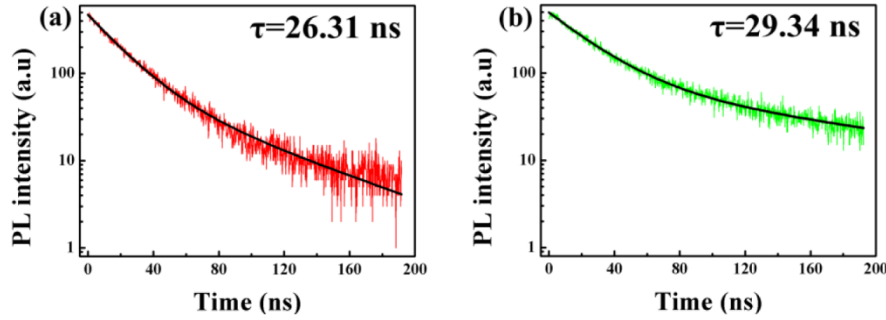


Fig. 9. Fluorescent lifetime of (a) red QDs; and (b) green QDs.

In this mathematical modeling, an ideal rectangular wave transferred by VLC system is analyzed. This rectangular wave signal can be observed in the oscilloscope since the photodetector receives this LED optical signal and transforms it into the current signal. The smearing produced by photodetector is far slighter than that of light conversion materials [18], so it can be ignored in our analyses. Then, the oscilloscope converts it into voltage signal and displays it. This voltage is given as:

$$U_{\text{signal}} = i \cdot r \quad (1)$$

where r is the internal resistance of oscilloscope and i is the current produced by photodetector, for the light with the wavelength of λ , i can be given by:

$$i = G \cdot M \cdot P_{\lambda} \cdot R_{\lambda} \quad (2)$$

where G is the transimpedance gain and M is a factor set to a constant. P_{λ} is the received optical power and R_{λ} is the responsivity of photodetector. On the other hand, P_{λ} can be obtained by the following function:

$$\Phi = K_m \cdot P_{\lambda} \cdot V_{\lambda} \quad (3)$$

where V_{λ} is the standard luminosity curve, K_m is the maximum visibility, and Φ is the luminous flux, which can also be expressed by light intensity by:

$$\Phi = \int I d\omega \quad (4)$$

where I is the light intensity and ω is the solid angle. Since the active area of photodetector is very small (about 1 mm²), the light intensity in this area can be regarded as a constant, therefore, i should be expressed as:

$$i = I \cdot \int_{\lambda_1}^{\lambda_2} \frac{\omega \cdot G \cdot M \cdot R_{\lambda}}{K_m \cdot V_{\lambda}} d\lambda \quad (5)$$

where λ_1 and λ_2 are determined by the wavelength range of LEDs. Therefore, Eq. (1) can be written as:

$$U_{\text{signal}} = I \cdot \int_{\lambda_1}^{\lambda_2} \frac{r \cdot \omega \cdot G \cdot M \cdot R_{\lambda}}{K_m \cdot V_{\lambda}} d\lambda \quad (6)$$

where the integral part is depended by instruments (including LED, photodetector and oscilloscope), so they can be regarded as an unchanging value when signals are being changed.

According to Eq. (6), it can be easily found that the signal will be obviously affected by light intensity. Fluorescent lifetime can be related with the $U_{smearing}$ of LED by as:

$$U_{smearing} = I_0 \exp(-t/\tau) \cdot \int_{\lambda_1}^{\lambda_2} \frac{r \cdot \omega \cdot G \cdot M \cdot R_{\lambda}}{K_m \cdot V_{\lambda}} d\lambda \quad (7)$$

where I_0 is the initial light intensity, τ is the fluorescent lifetime of light conversion material in LED and $U_{smearing}$ is the corresponding to the smearing process.

In general, for VLC system, light conversion materials of LED can be regarded as a linear time invariant system (LTI system). According to the Eq. (7), a longer τ leads to a longer time that $U_{smearing}$ occupies, then this longer smearing time, which can be regarded as a longer impulse response of this LTI system, corresponds a narrower frequency response in frequency domain. Furthermore, bandwidth can be represented by frequency response, namely, narrower frequency response leads to narrower bandwidth. Therefore, we can draw a conclusion that the longer the fluorescent lifetime is, the narrower the bandwidth is.

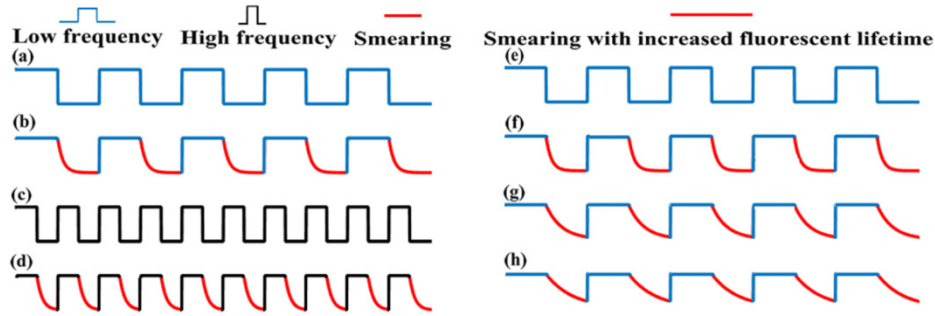


Fig. 10. The principles of smearing: (a) low frequency signal; (b) low frequency signal with smearing; (c) high frequency signal; (d) high frequency signal with smearing; (e)-(h) signal with smearing that more and more severe.

The smearing is showed in Fig. 10 with red lines. As shown in Fig. 10(a), ideal rectangular wave is transmitted. For WLED, due to the limitation of light conversion material's fluorescent lifetime, it requires some time for the signal from high voltage to low voltage and smearing is generated. Consequently, the wave form will be affected just as Fig. 10(b) shows. Furthermore, according to Fig. 10(a)-10(d), it can be easily found that for the same fluorescent lifetime, the higher the frequency of signal is, the severer the smearing is, thus, the signal distortion is more serious and the bandwidth is narrower. On the other hand, as shown in Fig. 10(e)-10(h), for the same signal frequency, the longer the fluorescent lifetime is, the longer the time from high voltage to low voltage needs, therefore, the signal distortion is more serious and the bandwidth is narrower.

According to these analyses, we can infer that QD-LED has a higher bandwidth than phosphor LED since phosphor has a far longer fluorescent lifetime. Furthermore, compared with the QD-LED based on green QDs, QD-LED based on red QDs with a shorter fluorescent lifetime, provides a higher bandwidth. Meanwhile, the QD-WLED, which consists of red and green QDs, will has a moderate bandwidth between that of those two types of QDs. Obviously, these predictions can be verified by Fig. 6. As for LMS, since it is formed by QDs and nonluminous microspheres, it will also have a higher bandwidth than traditional phosphor WLED, which can be verified by Fig. (8). Nevertheless, defects in LMS as well as light scattering caused by mesoporous in LMS may increase the time that smearing process occupied, so the bandwidth of QD-LED based on LMS is less than those based on QDs solution. In addition, it can be predicted that the higher the signal frequency is, the better the

performance of QD-LED will be. This characteristic can also be confirmed by Fig. 6 and Fig. 8, which indicates the noise of QD-LED is far lower than phosphor LED when the frequency is high.

4. Conclusion

In this study, the effects of QDs solution and QDs composites on improving the modulation bandwidths of LEDs were studied. Compared with the traditional LED based on YAG:Ce phosphor, the proposed QD-LED and QD-WLED reached maximum improvements on modulation bandwidths at 74.19% and 67.75% respectively. This result indicated the potential advantages of utilizing QD-LEDs in VLC, including good optical properties of QDs, and especially, higher transmission rate due to the wider bandwidth. Furthermore, the mechanism that how the fluorescent lifetime affects the bandwidth of QD-LEDs was also discussed. Due to the smearing, shorter fluorescent lifetime will lead to higher modulation bandwidth. Our findings can help build up the VLC system with better performance in lighting as well as in signals transmission.

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